PMARIS Asset Reliability Assessment

What 5,000 asset-health observations across six industrial sites say about where the next Egyptian pound of maintenance budget should go.

THE ANSWER

Put the next EGP on four controllable levers and on a wider escalation list. Loss is concentrated: the top 10% of observation windows carry **44.7%** of revenue loss. That concentration tracks things management controls (maintenance strategy, spare-parts availability, crew skill and PM discipline), each splitting downtime by roughly 2x to 3x. Meanwhile the executive priority flag misses **829** of the **960** high-risk windows.

1.88B	**11.0 h**	**19.2%**	**46.7%**	**829**
EGP revenue loss, 5,000 windows	mean unplanned downtime / window	windows high-risk (FP ≥ 50%)	of loss in those windows	high-risk, only P3/P4

The file at a glance

Property	Value
Source	Group_9230_Team1_PMARIS_Enterprise_Dataset.csv, one flat CSV
Shape	5,000 rows × 60 columns (41 numeric, 16 categorical, 3 identifiers)
Grain	One row = one 30-day observation window of one asset
Coverage	2025-09-01 to 2026-08-31, 12 months, 357–443 rows per month
Scope	6 sites in 4 industries: cement, oil & gas drilling, lab-equipment manufacturing, in-vitro diagnostics
Playbook path	Path A: single flat table with a transaction-level date column
Quality	0 missing cells, 0 duplicates, 0 type errors. One broken key (Asset_ID), see page 3

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EXECUTIVE SUMMARY

The money is concentrated, the drivers are controllable, and the escalation list is too narrow

Across 5,000 thirty-day observation windows, the assets lost **EGP 1.88B** of revenue to **54,938** hours of unplanned downtime. That loss is not spread evenly. Three arguments, each independently sufficient, support the answer on page 1.

1. Loss is concentrated where risk is high

The top 10% of windows carry 44.7% of revenue loss (Gini 0.63). The 960 windows with failure probability of 50% or more (19.2% of the file) carry 46.7% of loss and 45.0% of downtime. Windows with health under 70 lose 9x what windows above 90 lose.

2. The concentration tracks levers management controls

Reactive maintenance averages 19.0 h of downtime per window against 8.4 h for predictive (2.3x). Spare-part shortages average 20.8 h against 7.2 h with stock available (2.9x). Basic crews run 1.8x expert crews, and windows below 80% PM compliance run 2.2x those above. All survive a Bonferroni-corrected test with non-overlapping 95% intervals.

3. The executive flag does not see most of it

Only 134 windows are flagged P1 or P2. 829 of the 960 high-risk windows sit at P3 or P4, carrying EGP 723.3M of financial exposure, and P1/P2 cover only 26.5% of the loss in the worst 10% of windows. The flag is a fixed banding of a composite index that weighs failure probability only moderately.

The questions this review was asked, answered

Question	Answer	Basis
Where should the next EGP of maintenance effort go?	**Levers + escalation**	Pages 4 to 7
Is moving off reactive maintenance worth it?	**Yes**	2.3x downtime gap, large effect
Should one site be prioritised over the others?	**No**	Site explains 1.3% of loss variance
Does manufacturer (OEM) matter?	**No**	p = 0.57, not significant
Does operating shift matter?	**No**	p = 0.49, not significant
Is there a trend or season to schedule against?	**No**	Slope +0.6% of mean per month
Can individual assets be tracked or replaced from this file?	**No**	Asset_ID is not a stable key

METHOD

How the assessment was run

The file was taken through the AIB902 Agentic Analytics Playbook (v4.0) in order: intake and classification, a cleaning checklist, a check that each column means what its name says, descriptive insights, a significance gate, and a scoping interview with the stakeholder before anything was built. Method comes before findings so the findings are not read as opinion.

Stage 1 cleaning checklist: passed

Check	Result	Action
Missing cells	0 of 300,000	**None needed**
Exact duplicate rows / duplicate IDs	0 / 0	**None needed**
Numeric type errors, unparseable dates	0 / 0	**None needed**
Values beyond 3 SD (revenue loss)	113 rows	Kept: a real heavy tail, not errors
Load factor above 100%	452 rows	Kept: overload is physically possible
Rare categorical levels (<1% of rows)	3 failure modes, 27 rows	Pooled as 'Other' for analysis
Columns piled at a bound	Efficiency index 53% at 100	Treated as capped, not measured

Do the columns mean what they say? One does not

Column	Finding	Consequence
Asset_ID	**499 of 500 IDs change site; 498 change criticality; age runs backwards for 497**	Not an asset key. Each row is an independent window. No per-asset history, churn or replacement view
Financial_Exposure_At_Risk_EGP	r = 0.999 with revenue loss	Derived; never add the two
Failure_Probability_Pct	r = -0.972 with health index	Mirror images; report one
Executive_Intervention_Priority	Fixed bands of the Maintenance Priority Index (P4 < 40, P3 40–60, P2 60–74, P1 79+)	A formula, not a judgement; can be re-tuned
Composite scores (risk, priority, RUL)	Inter-correlated at 0.7–0.99	Describe a formula; not treated as discoveries

The significance gate

Every difference reported in this document was tested before it was described. 33 tests (11 dimensions × 3 measures) were run with one-way ANOVA and confirmed by the rank-based Kruskal-Wallis test; a difference counts only if both clear the Bonferroni threshold of 0.05 / 33 = 0.0015. 26 survive. With 5,000 rows almost any difference is detectable, so effect size (η^2, the share of variance explained) is reported beside every p-value and drives the wording: a real but small effect is called small.

FINDING 1

Loss is concentrated in the high-risk windows

A tenth of the windows carry 44.7% of revenue loss, and a fifth carry 64.8%. A Gini coefficient of 0.63 is in the range where prioritisation pays: effort spread evenly across the fleet is mostly spent where little is lost. 656 windows (13.1%) lost nothing at all.

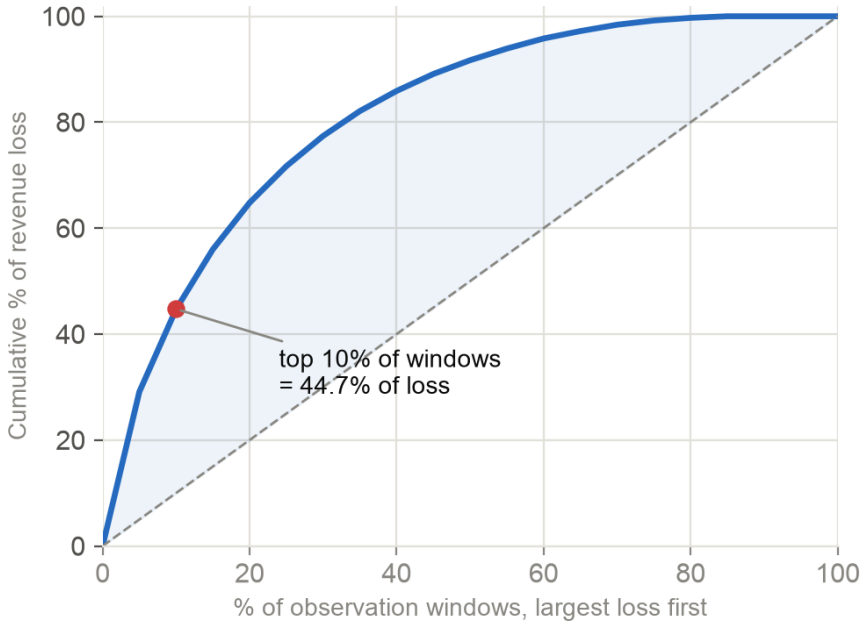


Figure 1. Lorenz curve of revenue loss across the 5,000 windows, largest first. The dashed diagonal is an even spread.

The concentration is not random: it follows asset health. Windows below a health index of 70 average EGP 1.1M of loss, 9x the EGP 125k of windows above 90, and run 31.7 h of downtime against 3.9 h.

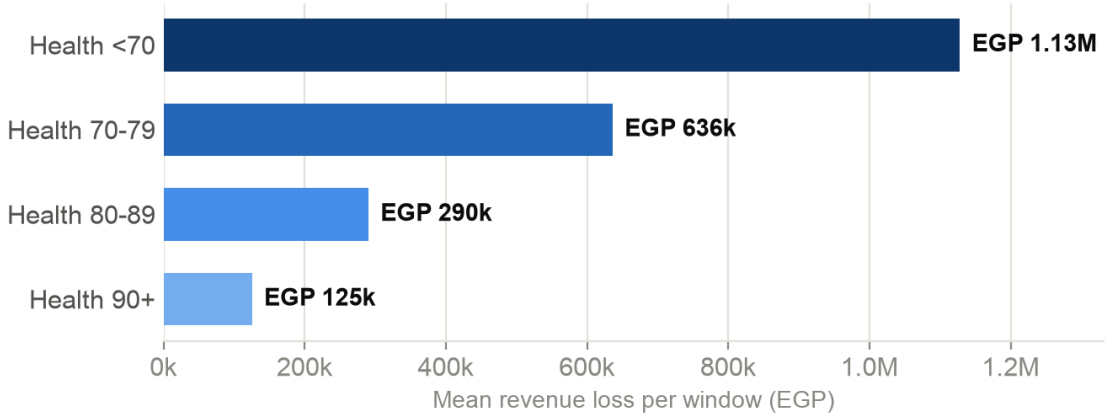


Figure 2. Mean revenue loss per window by Asset Health Index band. Live version: dashboard, Levers page.

SO WHAT

High-risk windows (failure probability ≥ 50%) are 19.2% of observations and carry 46.7% of revenue loss. Targeting them is the shortest path to the money.

FINDING 2

Maintenance strategy and spare-parts availability are the two biggest levers

Reactive windows average **19.0 h** of unplanned downtime; every planned strategy sits between 8.4 and 9.5 h, and their intervals overlap. The step that matters is leaving reactive, not choosing between the planned strategies. Reactive windows also run 56% emergency work orders against 23% for predictive (η^2 0.076 on downtime, p = 4e-85).

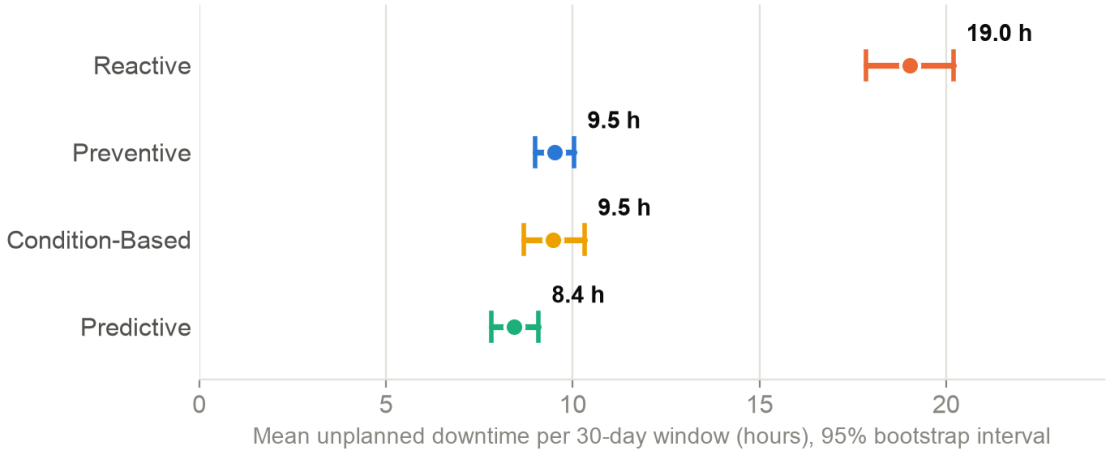


Figure 3. Mean downtime by maintenance strategy with 95% bootstrap intervals (2,000 resamples).

Spare parts show a clean staircase: shortage 20.8 h, limited 13.2 h, available 7.2 h, with no interval touching the next (η^2 0.091). Spares status is the single strongest categorical split on downtime in the file.

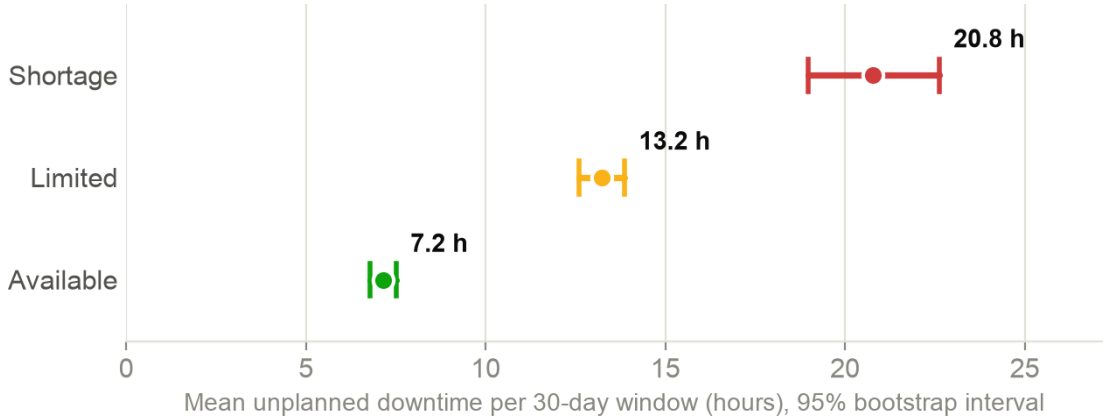


Figure 4. Mean downtime by spare-parts status with 95% bootstrap intervals.

> **ESTIMATED, NOT MEASURED**
> Illustrative sizing, assuming the association were fully causal: had the 906 reactive windows run at the preventive average, downtime would have been 8,621 h lower (16% of the total). Had the 426 shortage windows run at the available-stock average, 5,800 h lower (11%). These are upper bounds, not forecasts: reactive windows also start in worse health.

FINDING 3

Crew skill and PM discipline make up the other two levers, and the levers compound

Basic crews average 14.8 h against 8.0 h for expert crews. PM compliance shows a steady dose-response: the least compliant decile averages 22.8 h, the most compliant 6.5 h.

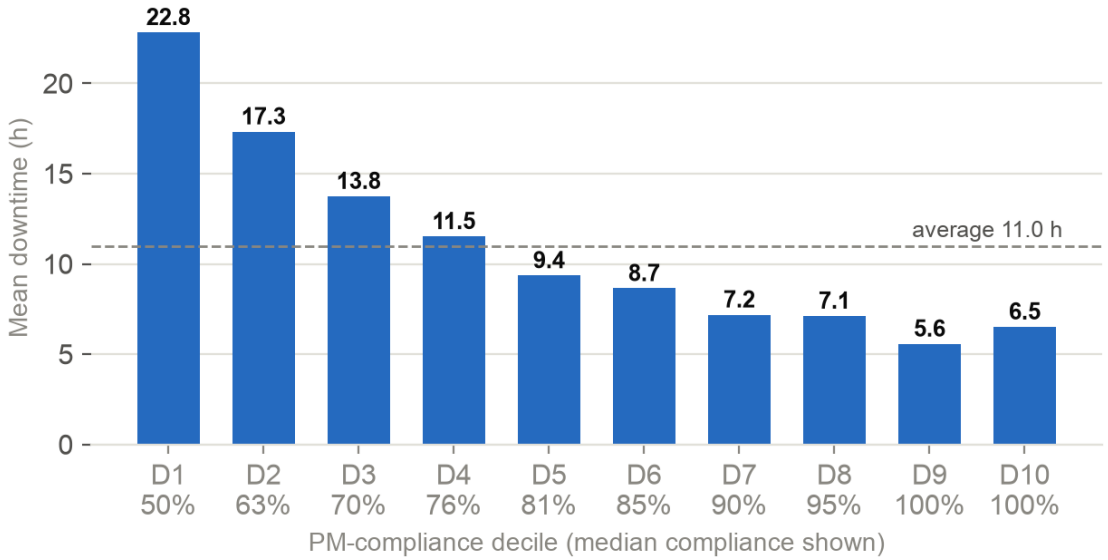


Figure 5. Mean downtime by PM-compliance decile (D1 least compliant).

Lever thresholds, below versus above

Lever	Worse side	Downtime	Better side	Downtime	Ratio	p	η²
PM compliance	below 80%	16.0 h	80% or more	7.3 h	2.2x	4e-113	0.097
Days since last maintenance	60+ days	15.7 h	under 60 days	8.7 h	1.8x	4e-65	0.056
Lubricant quality index	LQI under 50	15.8 h	LQI 50+	7.9 h	2.0x	7e-90	0.078
Maintenance backlog	backlog 14d+	15.5 h	backlog under 14d	7.7 h	2.0x	1e-89	0.078

The levers compound. The worst combination, reactive strategy with a parts shortage, averages 26.3 h of downtime; predictive with parts available averages 5.9 h, a 4.5x spread.

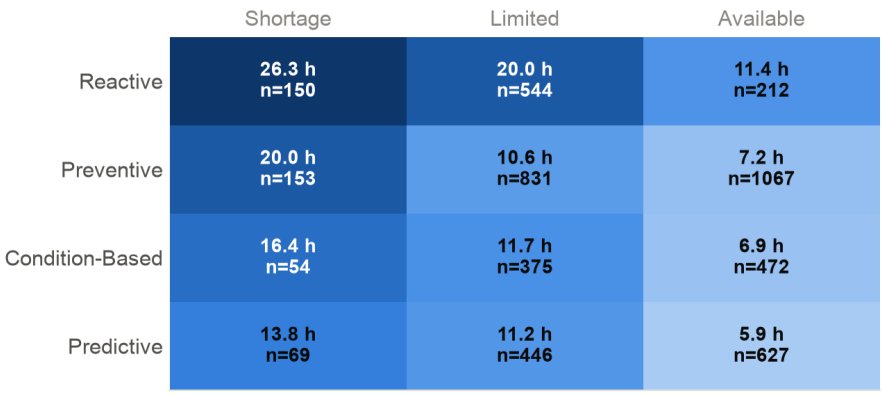


Figure 6. Mean downtime by strategy × spare-parts status, with observation counts per cell.

FINDING 4

The executive escalation flag misses most of the risk

Executive Intervention Priority is a fixed banding of the Maintenance Priority Index. That index correlates only 0.75 with failure probability, so high-risk windows routinely land in P3 or P4. Of the 960 windows with failure probability of 50% or more, **829** are not escalated. They carry **EGP 723.3M** of financial exposure, 32.6% of the file's total.

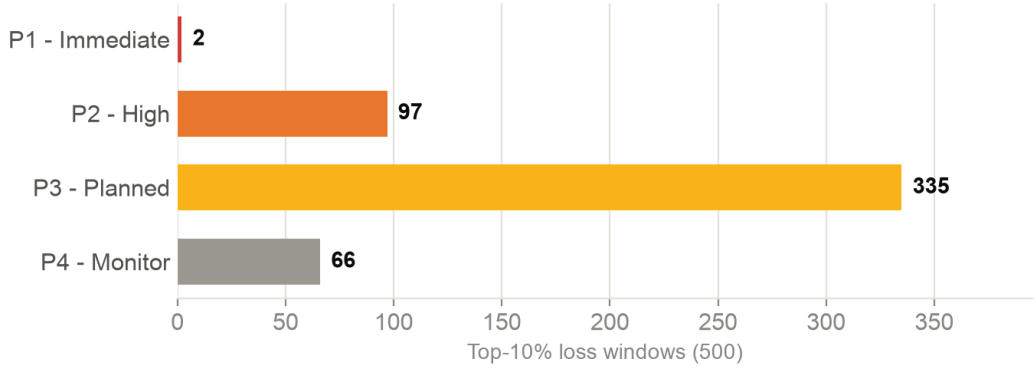


Figure 7. The 500 windows with the largest revenue loss, by the flag they received. P1/P2 cover 26.5% of their loss.

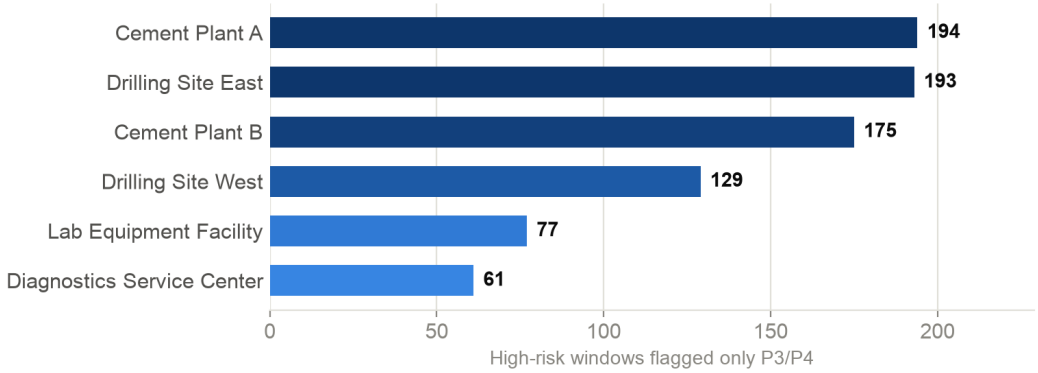


Figure 8. Unescalated high-risk windows by site. Every site has them; heavy-industry sites have the most.

RECOMMENDED RULE

Add a second escalation trigger: failure probability ≥ 50% and criticality High or Critical. On this file it would add **433** windows carrying **EGP 434.3M** of exposure (60% of the gap's exposure). It would raise the escalated count from 134 to 567, so capacity to act on it must be confirmed before it goes live.

CONTEXT

Where loss sits, and what does not matter

Cement and drilling sites lose more per window than the lab and diagnostics sites, and the gap is real (p = 7e-13). It is also small: site explains 1.3% of loss variance. Criticality matters more for money: critical windows average EGP 597k, 4.4x low-criticality ones (η^2 0.064).

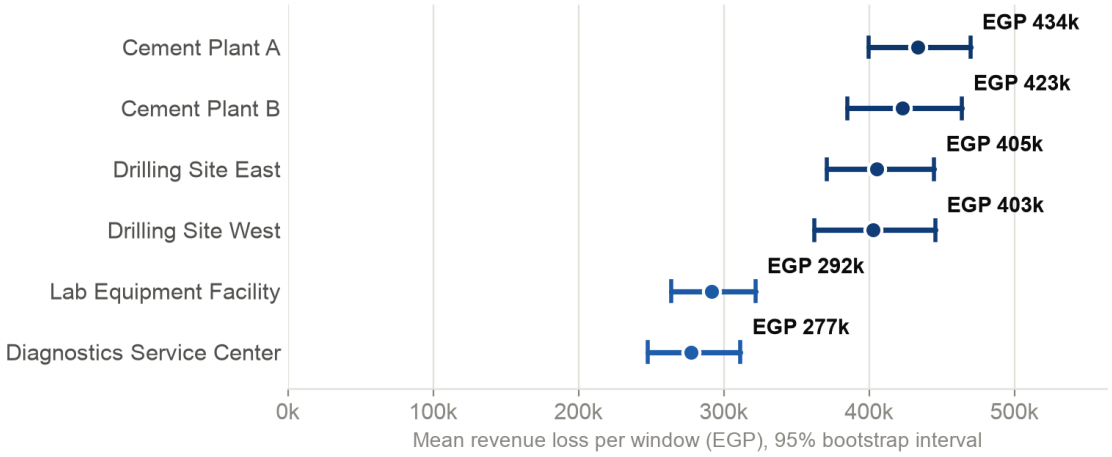


Figure 9. Mean revenue loss by site with 95% bootstrap intervals. The four heavy-industry sites overlap each other.

What was tested and found to carry no signal

Dimension	Downtime p	Loss p	Failure-prob. p	Verdict
Manufacturer Group	0.57	0.75	0.90	No difference
Operating Shift	0.49	0.01	0.93	No difference
Month (trend)	–	slope +0.60%/mo	–	Flat; lag-1 r = -0.17

A site-versus-site campaign, an OEM switch or a shift change would be acting on a difference the data does not show. Environment severity does matter for risk (severe environments are 3.8x as likely to be high-risk as controlled ones), but environment is a condition to plan around, not a lever.

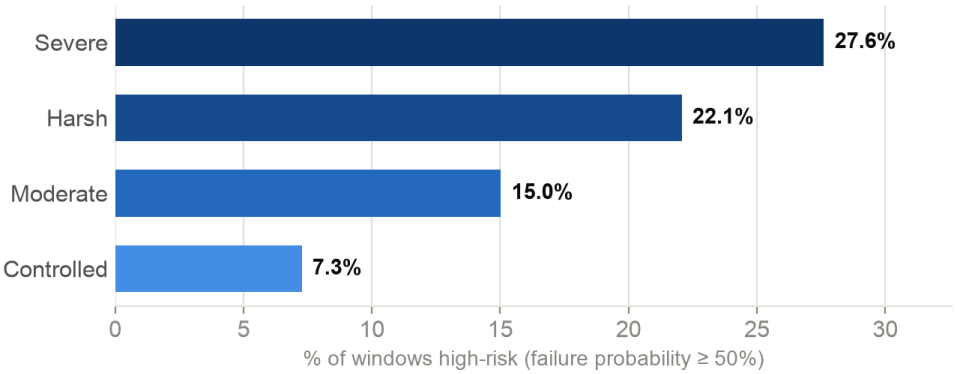


Figure 10. Share of windows that are high-risk, by environment severity.

EVIDENCE

The significance register, and the limits of this file

All 33 tests, ranked by effect size. Both tests must clear p < 0.0015.

Dimension	Measure	F	p (ANOVA)	p (K-W)	η^2	Verdict
Failure Mode Group	Failure prob.	971.6	<1e-300	<1e-300	0.3684	**Real, Large**
Maintenance Strategy	Failure prob.	396.0	9e-231	2e-174	0.1921	**Real, Large**
Spare Parts Status	Failure prob.	410.4	1e-165	5e-148	0.1411	**Real, Large**
Failure Mode Group	Downtime	210.5	1e-128	7e-129	0.1122	**Real, Medium**
Failure Mode Group	Revenue loss	169.3	2e-104	3e-108	0.0923	**Real, Medium**
Spare Parts Status	Downtime	249.4	6e-104	2e-115	0.0908	**Real, Medium**
Maintenance Strategy	Downtime	136.7	4e-85	9e-80	0.0758	**Real, Medium**
Environment Severity	Failure prob.	119.8	6e-75	6e-88	0.0671	**Real, Medium**
Spare Parts Status	Revenue loss	174.0	9e-74	9e-76	0.0651	**Real, Medium**
Criticality Class	Revenue loss	113.7	3e-71	3e-83	0.0639	**Real, Medium**
Site	Failure prob.	49.8	2e-50	4e-59	0.0475	**Real, Small**
Industry Context	Failure prob.	82.2	7e-52	1e-60	0.0470	**Real, Small**
Maintenance Strategy	Revenue loss	81.9	1e-51	9e-57	0.0469	**Real, Small**
Asset Class	Failure prob.	20.2	9e-34	6e-40	0.0352	**Real, Small**
Technician Skill Level	Failure prob.	42.9	2e-27	6e-28	0.0251	**Real, Small**
Technician Skill Level	Downtime	41.4	2e-26	1e-23	0.0243	**Real, Small**
Environment Severity	Downtime	32.0	2e-20	1e-25	0.0188	**Real, Small**
Technician Skill Level	Revenue loss	28.1	5e-18	1e-19	0.0166	**Real, Small**
Environment Severity	Revenue loss	27.0	3e-17	3e-22	0.0159	**Real, Small**
Site	Downtime	15.2	7e-15	5e-21	0.0150	**Real, Small**
Industry Context	Downtime	25.1	4e-16	4e-22	0.0148	**Real, Small**
Site	Revenue loss	13.3	7e-13	1e-19	0.0131	**Real, Small**
Industry Context	Revenue loss	22.1	3e-14	5e-21	0.0131	**Real, Small**
Asset Class	Downtime	6.7	1e-09	9e-15	0.0119	**Real, Small**
Asset Class	Revenue loss	6.1	2e-08	2e-13	0.0109	**Real, Small**
Criticality Class	Failure prob.	14.5	2e-09	2e-10	0.0086	**Real But Negligible**
Criticality Class	Downtime	3.5	0.015	7e-05	0.0021	**Not Significant**
Operating Shift	Revenue loss	4.3	0.013	0.843	0.0017	**Not Significant**
Manufacturer Group	Downtime	0.8	0.567	0.564	0.0008	**Not Significant**
Manufacturer Group	Revenue loss	0.5	0.747	0.297	0.0005	**Not Significant**
Manufacturer Group	Failure prob.	0.3	0.899	0.963	0.0003	**Not Significant**
Operating Shift	Downtime	0.7	0.486	0.885	0.0003	**Not Significant**
Operating Shift	Failure prob.	0.1	0.932	0.864	0.0000	**Not Significant**

Limits a reader must keep in mind

- **Association, not proof of cause.** This is observational data. Reactive windows start in worse health, so part of the strategy gap is which assets end up reactive. A controlled pilot is the way to measure the causal share.
- **No asset history.** Asset_ID is not a stable key, so nothing here tracks one machine over time, and no per-asset replacement, remaining-life or churn view is supportable.
- **Summed 30-day windows are not annual totals.** Totals add overlapping 30-day snapshots; use means per window for comparisons.

RECOMMENDATIONS

What to do, what may be published, and how to reproduce it

#	Recommendation	Evidence
1	Move reactive windows onto a planned strategy, starting with critical and high-criticality assets.	906 reactive windows average 19.0 h vs 9.5 h preventive.
2	Fund spare-parts availability for the asset classes that hit shortage.	Shortage 20.8 h vs available 7.2 h; the largest categorical effect.
3	Hold PM compliance at 80% or above, and clear backlog beyond 14 days.	Each threshold splits downtime about 2x.
4	Prioritise upskilling basic and intermediate crews on high-risk assets.	Basic 14.8 h vs expert 8.0 h.
5	Add the second escalation trigger (FP ≥ 50% and criticality High/Critical), after confirming capacity.	Adds 433 windows and EGP 434.3M of exposure to the watched list.
6	Run a controlled pilot before scaling: convert one site's reactive fleet and measure against a matched site.	Turns an association into a causal estimate before full budget is committed.
7	Fix the asset key in the next export.	A stable Asset_ID unlocks per-asset history, remaining-life tracking and replacement planning.

What may be published

Safe to state	Do not state
Lever differences (strategy, spares, skill, PM) with their sizes	That any lever causes the reduction, until a pilot measures it
Criticality and environment as real risk factors	A ranking of sites as better or worse managed
Concentration figures and the escalation gap	Anything about manufacturer or shift quality
Means per 30-day window	Per-asset histories, remaining life by machine, or annual totals

Method and reproducibility

- **Analysis:** `python 02_Analysis/analyze_pmaris.py`, standard library only, about 90 s; writes findings.json.
- **Figures and report:** `python 04_Deliverables/make_figures.py` then `make_report.py` (matplotlib, reportlab).
- **Live dashboard:** `python watcher.py` and `python -m http.server 8000` in 03_Dashboard, six pages. Live refresh measured at 376 ms mean, 621 ms max over 10 timed edits.
- **Measured vs estimated:** every figure is measured from the file except the two sizing estimates on page 5, which are labelled.